

Manuel Pichardo Marcano

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Pronouns: he/him/his

Research Interests

Compact Objects, Compact Objects in Globular Clusters, White Dwarf Binaries, Cataclysmic Variables, AM CVns, Variable Stars, Binary systems, Data analysis, Big data, Citizen Science, Outreach.

Postdoctoral Research Experience

- ▶ *Instituto de Astronomía, Universidad Nacional Autónoma de México. Ciudad de México, México*
Postdoctoral Fellow
September 2025 - Present
- ▶ *Fisk/Vanderbilt University, Nashville, TN*
EMIT (Establishing Multimessenger Astronomy Inclusive Training) Postdoctoral Fellow
September 2023 - September 2025
- ▶ *American Museum of Natural History, New York, NY*
Kalbfleisch Postdoctoral Research Fellow
September 2022 - September 2023

Education

- ▶ *PhD., Physics*
Texas Tech University, 2018-2022
Project: Variability of Accreting White Dwarf Binaries and Other Compact Objects
Advisors: Professor Thomas J. Maccarone and Professor Liliana E. Rivera Sandoval
- ▶ *Master of Science in Physics*
Texas Tech University, 2016-2018
- ▶ *Joint European Master Degree in Space Science & Technology*
Project: MUSE observations of the compact objects NGC 6397
M.S., Space and Science Technology, Luleå tekniska universitet (2014 - 2016)
M.S., Astrophysics, Université Paul Sabatier Toulouse III, Mention Assez Bien (2014 - 2016)
Advisors: Dr. Natalie Webb and Dr. Sebastien Guillot
- ▶ *Bachelor of Science, Physics*
Utah State University, Cum Laude (2014)

Publications

Papers

First author

- ▶ **Pichardo Marcano, M.**, Maccarone, T. J., & Rivera Sandoval, L. E. (2026). A 682-second X-ray periodicity in CH Cygni: evidence for a magnetic white dwarf, *Monthly Notices of the Royal Astronomical Society*, 546, stag299.

- **Pichardo Marcano, M.**, Yelikar, A. B., & Jani, K. (2025). Massive Double White Dwarf Binary Mergers from the Moon: Extending the Reach of Multimessenger Astrophysics, *The Astrophysical Journal*, 992, 16.
- **Pichardo Marcano, M.**, Rivera Sandoval, L. E., Maccarone, T. J., Rohrmann, R. D., Althaus, L. G., Heinke, C. O., Belloni, D., & Bahramian, A. (2025). A Second Candidate Magnetic Helium Core White Dwarf and 3 Other Variable White Dwarfs in the Globular Cluster NGC 6397, *The Astrophysical Journal*, 979, 167.
- **Pichardo Marcano, M.**, Rivera Sandoval, L. E., Maccarone, T. J., Rohrmann, R. D., Heinke, C. O., Belloni, D., Althaus, L. G., & Bahramian, A. (2023). A candidate magnetic helium-core white dwarf in the globular cluster NGC 6397, *Monthly Notices of the Royal Astronomical Society*, 521, 5026.
- **Pichardo Marcano, M.**, Rivera Sandoval, L. E., Maccarone, T. J., & Scaringi, S. (2021). TACOS: TESS AM CVn Outbursts Survey, *Monthly Notices of the Royal Astronomical Society*, 508, 3275.
- **Pichardo Marcano, M.**, Rivera Sandoval, L. E., Maccarone, T. J., Zhao, Y., & Heinke, C. O. (2021). A 2-d orbital period for a redback millisecond pulsar candidate in the globular cluster NGC 6397, *Monthly Notices of the Royal Astronomical Society*, 503, L51.

Co-author

- Salazar Manzano, L. E., Rivera Sandoval, L. E., Hameury, J.-M., Heinke, C. O., Kotko, I., Maccarone, T. J., & **Pichardo Marcano, M.** (2026). KL Dra as a Benchmark Laboratory for Accretion-Disk Physics: Constraints from TESS and Ground-Based Surveys, (Accepted in PASA), arXiv:2601.04323.
- Dag, M. K., Scaringi, S., [...], **Pichardo Marcano, M.** et al. (2026). A catalogue of orbital periods of cataclysmic variables and candidates from TESS observations, *Monthly Notices of the Royal Astronomical Society*, 548, stag514.
- Kára, J., Rivera Sandoval, L., Mendoza, W., Maccarone, T. J., **Pichardo Marcano, M.**, Salazar Manzano, L. E., Oelkers, R. J., & van Roestel, J. (2025). A study of transients from ground-based surveys reveals new AM CVn stars, (Accepted in PASA), arXiv:2511.16774
- Khandelwal, A., Das, S. N., Sharma, [...], **Pichardo Marcano, M.**, et al. (2026). TOI-6884b: A low-mass brown dwarf transiting a slightly evolved star, (Accepted in MNRAS).
- Gomez Maqueo Chew, Y., Dransfield, G., Barkaoui, K., [...], **Pichardo Marcano, M.**, Plauchu-Frayn, I et al. (2026). TOI-1080 b: a temperate, rocky planet orbiting a quiet M4V host, *Monthly Notices of the Royal Astronomical Society*, 548, stag438.
- Peláez-Torres, A., Pozuelos, F. J., Morello, G., DĀ©vora-Pajares, [...], **Pichardo Marcano, M.**, Plauchu-Frayn, I., Queloz, D. et al (2026). A gem system with a lava world and a habitable zone sub-Neptune orbiting TOI-1752, *Monthly Notices of the Royal Astronomical Society*.
- Morello, G., Peláez-Torres, A., Pozuelos, F. J., [...], **Pichardo Marcano, M.**, Plauchu-Frayn, I., Queloz, D. et al. (2026). The K-dwarfs Survey I. Four Validated Planets from the Radius Valley to the Neptune Desert, *Monthly Notices of the Royal Astronomical Society*.
- Pattie, E. C., Maccarone, T. J., Tetarenko, A. J., Miller-Jones, J. C. A., **Pichardo Marcano, M.**, & Rivera Sandoval, L. E. (2024). Variable Radio Emission of Neutron Star X-ray Binary Ser X-1 during Its Persistent Soft State, *The Astrophysical Journal*, 970, 136.
- Schwöpe, A., Buckley, D. A. H., Kawka, A., König, O., Lutovinov, A., Maitra, C., Mereminskiy, I., Miller-Jones, J., **Pichardo Marcano, M.**, Rau, A., Semena, A., Townsend, L. J., & Wilms, J. (2021). Identification of SRGt 062340.2-265715 as a bright, strongly variable, novalike cataclysmic variable, arXiv e-prints, arXiv:2106.14538.
- Rivera Sandoval, L. E., Maccarone, T. J., & **Pichardo Marcano, M.** (2020). A Year-long Superoutburst from an Ultra-compact White Dwarf Binary Reveals the Importance of Donor Star Irradiation, *The Astrophysical Journal*, 900, L37.
- Zhao, Y., Heinke, C. O., Tudor, V., Bahramian, A., Miller-Jones, J. C. A., Sivakoff, G. R., Strader, J., Chomiuk, L., Shishkovsky, L., Maccarone, T. J., **Pichardo Marcano, M.**, & Gelfand, J. D. (2020). The MAVERIC survey: a hidden pulsar and a black hole candidate in ATCA radio imaging of the globular cluster NGC 6397, *Monthly Notices of the Royal Astronomical Society*, 493, 6033.

- S.W. McIntosh, W.J. Cramer, **M. Pichardo Marcano**, R.J. Leamon. *The detection of Rossby-like waves on the Sun*. Nature Astron. 1, 0086 (2017). doi:10.1038/s41550-017-0086

White Papers and Pre-prints

- Jani, K., Abernathy, M., [...], **Pichardo Marcano, M.**, Panning, M., do Vale Pereira, P., et al (2025). Laser Interferometer Lunar Antenna (LILA): Advancing the U.S. Priorities in Gravitational-wave and Lunar Science, arXiv e-prints, arXiv:2508.11631.
- Lang, F. Z., Demory, B. O., Gomez Maqueo Chew, Y., Schmid, Y., Timmermans, M., [...], **Pichardo Marcano, M.**, & Hooton, M. J. (2026). TOI-4616 b: a benchmark Earth-sized planet transiting a nearby M4 dwarf, arXiv e-prints, arXiv:2603.10905.
- Macrie, C. W., Rivera Sandoval, L., Cavecchi, Y., Wong, T. L. S., & **Pichardo Marcano, M.** (2024), Research Notes of the American Astronomical Society, 8, 299.
- Maccarone, T. J., Beardmore, A., Mukai, K., Page, K., **Pichardo Marcano, M.**, & Rivera Sandoval, L. (2021). X-ray Pulsations from Nova Her 2021, The Astronomer's Telegram, 14776, 1.
- **Pichardo Marcano, M.** (2020). A period of 3.9 hours in the TESS light curve of SRGt 062340.2-265715, The Astronomer's Telegram, 14222, 1.
- Jeffrey S. Hazboun, **Manuel Pichardo Marcano** and Shane L. Larson. *Limiting alternative theories of gravity using gravitational wave observations*
preprint <http://arxiv.org/abs/1311.3153>

Talks and Posters

- *White Dwarf Binaries: From Outbursts to Gravitational Waves*
Talk. Coloquio Arcadio Poveda. UNAM. Mexico City. April. 2026.
- *Lunar Gravitational-Wave Detectors: Enabling Multi-Messenger Astronomy of White Dwarf Mergers with LSST*
Talk. Seminario MAPTA-UANL. Feb. 2025.
- *Lunar Gravitational-Wave Detectors: Enabling Multi-Messenger Astronomy of White Dwarf Mergers with LSST*
Talk. Second Annual LSST Latin American Meeting. Dec. 2025.
- *Science Explorer: Not Your Adviser's ADS*
Talk. UTRGV Astronomy Journal Club. Feb. 2025.
- *Seeing the most massive white dwarf mergers with LILA*
Talk. 245th AAS Meeting. 2025.
- *From ADS to SciX: Extending ADS to all NASA Science*
Talk. Vanderbilt Astronomy Seminar Series. August 2024.
- *Variability of Accreting White Dwarf Binaries*
Talk. V International Congress of Dominican Society of Physics. 2024.
- *MARES Survey: Magnetic Remnants in Symbiotics*
Poster. XVII Latin American Regional IAU Meeting. 2023.
- *SCOVaS: Survey for Compact Objects and Variables Stars*
Talk. Two in a Million - The interplay of binaries and star clusters. 2023
- *SCOVaS: Survey for Compact Objects and Variables Stars*
Talk. The Transient and Variable Universe. 2023
- *Variability of accreting white dwarf binaries and other compact objects* (Invited)
Lunch Talk. Carnegie Observatories. 2023

- ▶ *A candidate magnetic helium core white dwarf binary in the globular cluster NGC 6397*
Talk. 241th AAS Meeting. 2023.
- ▶ *A search for variable stars and compact binaries in globular clusters with HST (Invited)*
Talk. Trinity University Physics Department Seminar. 2022
- ▶ *TACOS: TESS AM CVn Outbursts Survey*
Talk. AM CVn 4.5 Meeting. September 2022.
- ▶ *Variability of Accreting White Dwarf Binaries and Other Compact Objects*
Dissertation Talk. 240th AAS Meeting. 2022.
- ▶ *A search for variable stars and compact binaries in globular clusters with HST (Invited)*
Talk. UTRGV Physics Department Colloquium. 2022
- ▶ *Results from the TESS AM CVn Outbursts Survey (TACOS)*
Talk. TESS Science Team Meeting #26. 2021.
- ▶ *Astronomía en Ultravioleta. (Invited)*
Talk. Alpha-Cen Summer School. 2021.
- ▶ *A Search for Variable Stars and Compact Binaries in Globular Clusters with HST*
Talk. Alpha-Cen Annual Meeting and General Assembly. 2020
- ▶ *SSCOVaS: Survey for Studying Compact Objects and Variables Stars*
Poster. 36th Annual New Mexico Symposium. 2020
- ▶ *"Compact Binaries in the globular cluster NGC 6397"*
Talk. Texas APS Section. Lubbock, TX. 2019
- ▶ *"MUSE integral field unit observations of the compact objects in the globular cluster NGC 6397"*
Poster. Stellar Remnants at the Junction. Junction, TX. 2016
- ▶ *"Measuring Water Level Fluctuations of Two Connected Wetlands in the Dominican Republic Using InSAR"*
Poster. SHPE National Conference. Dallas, TX. 2012
- ▶ *"Measuring Water Level Fluctuations of Two Connected Wetlands in the Dominican Republic Using InSAR"*
Talk. USU Physics Department Colloquium. 2012

Proposals

- ▶ *OPTICAM: Multi-band transit follow-up of giant planet candidates around M-dwarfs. Cycle 2026A. 8 nights at the 2.1 meter National Astronomical Observatory of Mexico (Co-Investigator)*
- ▶ *NEID Spectrograph: Mass Measurements of Two Gas Giant Planets around M-dwarf stars. 1.8 night. (Co-Investigator)*
- ▶ *Transiting Exoplanet Survey Satellite (TESS): "COPAS: Compact Objects And The Physics Of Accretion Survey" Large Program. (Co-Investigator)*
- ▶ *The Gemini Observatory: Cycle 2023B (16 hrs). (Co-Investigator)*
- ▶ *Hubble Space Telescope (HST): Archival research grant proposal. A search for variable stars and compact binaries in globular clusters with HST (Co-Investigator)*
- ▶ *Transiting Exoplanet Survey Satellite (TESS): "Studying the outbursts of AM CVns with TESS" (Co-Investigator)*
- ▶ *Multi Unit Spectroscopic Explorer (MUSE): The Hydrogen Alpha Disk Emission from the Compact Binaries of the Globular Cluster NGC 6397 (Co-Investigator 1.5 h)*

Teaching and Mentoring Experience

Teaching

- ▶ Professor, Mathematical Methods for Physics: Instructor of record for graduate-level mathematical methods in the Master's Program in Physics Education. Autonomous University of Santo Domingo (UASD), Azua Campus, Spring 2025.
- ▶ Graduate Part-Time Instructor PHYS 1403/1404: Instructor of record for Algebra-based, Inquiry-based, Laboratory-based Introductory Physics. TTU, 2017.
- ▶ Graduate Part-Time Instructor PHYS 1403/1404: Instructor of record for Algebra-based, Inquiry-based, Laboratory-based Introductory Physics. TTU, 2017.
- ▶ Teaching Assistant PHYS 1401/2401: Principles of Physics I/II. Lead the laboratories and discussions for the calculus based introductory physics. TTU, 2016-2017.
- ▶ Undergraduate Teaching Fellow PHYS 2215: Physics for Scientists and Engineers Lab I - Utah State, Fall 2013.

Mentoring

- ▶ Research advisor and Co-founder of the Fisk-Vanderbilt IDEA Summer Scholars Program. Mentored one undergraduate student from Fisk University in a summer long program. Summer 2025.
for the *Women and girls in astronomy program*. Finding variable White Dwarfs in Old Open Clusters. Fall 2024.
- ▶ Research advisor for the *Women and girls in astronomy program*. Finding variable White Dwarfs in Old Open Clusters. Fall 2024.
- ▶ Research advisor for *The Central American-Caribbean bridge in astrophysics (CENCA)*. CENCA is a remote research experience for undergraduates. Fall 2024.
- ▶ *Fisk University*: Mentor 3 groups of computer science students on an Astronomy Data Science project. Spring 2024.
- ▶ Mentor for *The Science Research Mentoring Program (SRMP)* at the American Museum of Natural History. Mentored 3 high school students to conduct research in machine learning in astronomy. Sept 2022 - June 2023
- ▶ Mentor for *Alpha-Cen: Astrophysics in Central America and the Caribbean*. Spring 2023.
- ▶ Mentor for *The Central American-Caribbean bridge in astrophysics (CENCA)*. CENCA is a remote research experience for undergraduates. Spring 2021, Fall 2022/23

Recognition & Leadership Experiences

- ▶ Co-Leader, Central American-Caribbean Bridge in Astrophysics (CENCA Bridge). 2025-Present
Simons Foundation (\$1.5M, 2025 to 2030) and IAU-OAD funded (2022 to 2024) nonprofit providing remote research internships for undergraduate students in Central America and the Caribbean. Recipients of the IAU Development Prize 2024.
- ▶ Ciencia Pal Barrio, co-founder and organizer. A monthly science talk at a local brewery in Ciudad de México. 2025-Present.
- ▶ Zooniverse Ambassador. 2025
Zooniverse is the world's largest platform for online participatory research.
- ▶ Science Explorer (SciX) Lead Ambassadors. 2024-2026
SciX is the expansion of the Astrophysics Data System (ADS) to all of NASA Science.
- ▶ Outstanding Ph.D. Student: *TTU Physics & Astronomy Department*. 2022
- ▶ FAMOUS (Funds for Astronomical Meetings: Outreach to Underrepresented Scientists) travel grant for the 239th meeting of the AAS.
- ▶ Co-organizer First ComSciCon en Español. The communicating science workshop for graduate students.

- ▶ Bucy Scholarship in Applied Physics: *TTU Physics & Astronomy Department. 2020-2021*
- ▶ Erasmus Mundus Scholarship: 2-year scholarship awarded by the European Commission
- ▶ Presidential Scholarship: Full 4-year scholarship awarded by the Dominican Government
- ▶ Member: *Sigma Pi Sigma*, The Physics Honor Society
- ▶ Lawrence R. and Abeline Megill Scholarship: *USU Physics Department*

Professional Service

- ▶ Proposal Reviewer: NSF NOIRLab 26A & 26B Galactic and Solar System TAC. 2025 & 2026
- ▶ Peer Reviewer: Astronomy & Astrophysics, Publications of the Astronomical Society of Japan. 2025

Technical Skills

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| <ul style="list-style-type: none"> ▶ <i>Programming Languages</i>: Proficient in Python (Jupyter/iPython notebooks) and shell scripting (bash). Basic knowledge of R, RStudio and Tidyverse packages. ▶ <i>Markup Languages</i>: $\text{\LaTeX}$, Markdown ▶ <i>Operating Systems</i>: GNU/Linux, OS X, Windows | <ul style="list-style-type: none"> ▶ <i>Software</i>: IRAF/PyRAF, DOLPHOT, SAOImageDS9/JS9 ▶ <i>Version Control Systems</i>: Git ▶ <i>Virtualization and Cloud Services</i>: Docker (basic) |
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Languages

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| <ul style="list-style-type: none"> ▶ <i>Spanish</i>: Native Language ▶ <i>English</i>: Fluent | <ul style="list-style-type: none"> ▶ <i>French</i>: Intermediate knowledge (B1) |
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Professional Development

Workshops

- ▶ **Summer School in Open Science + Research Software Engineering**: One-week summer school with Lectures on open science and software development best practices. July 2024.
- ▶ **National Data Science Alliance-Research Affinity Cohorts**: Two-week summer research project. I worked with Professor Deborah Sunter on a project evaluating equitable energy access in the United States. Clark Atlanta University. June, 2024
- ▶ **Spring 2024 National Data Science Alliance Workshop**: Exploratory Data Analysis (EDA) & Data Visualization. March, 2024
- ▶ **Spring 2024 National Data Science Alliance Workshop**: Navigating the Data Science Landscape: The What, Why, and How? January, 2024
- ▶ **EMIT Summer School in Multimessenger Astronomy**: Lectures and tutorials in gravitational waves, statistics, multimessenger astronomy, STEM education and diversity and inclusion topics. Summer, 2023
- ▶ **Penn State Summer School in Statistics for Astronomers**: Lectures and tutorials in statistics, astronomy, computer science, and informatics. Summer, 2021
- ▶ **SciPy 2021**: Scientific Computing with Python conference. Summer, 2021
- ▶ **Wiki Scientist Course**: Elevating the Visibility of Quantum Scientists on Wikipedia by the American Physical Society and Wiki Education. Spring, 2021. Blog post: Improving the quality of Spanish language articles on Wikipedia

- ▶ **ComSciCon:** The communicating science workshop for graduate students. June, 2020.
- ▶ **Reclaiming STEM:** Diverse & inclusive SciComm workshop. Fall, 2020.
- ▶ **JWST proposal preparation workshop:** Instrument description, planning tools and hands-on sessions. Spring, 2020.
- ▶ **NRAO Community Days at TTU:** Two-day event, including hands-on sessions and observation preparation and data reduction. Spring, 2019.

Public Outreach and Community Service

- ▶ **Ciencia Pal Barrio**, co-founder and organizer. A monthly science talk at a local brewery in Ciudad de México. 2025–Present.
- ▶ *Past Author at Astrobites*: One of the original authors. Astrobites is the Spanish language version of Astrobites. URL: <https://astrobites.org/author/manuelmarcano22/>
- ▶ *Science Explorer (SciX) Ambassador*
SciX is the expansion of the Astrophysics Data System (ADS) to all of NASA Science. Blog Post. *From ADS to SciX: What I wish I knew about ADS/SciX during my Ph.D.*
- ▶ *Letters to a Pre-Scientist*, pen pal. 2025–2026.
- ▶ *Part of the team translating to Spanish the "Machine Learning for Physics and Astronomy" course*. https://openlearning.flatironinstitute.org/courses/course-v1:cca+ML_01+A/about

Talks

- ▶ *A Needle in a Haystack: Finding Exotic Binaries in Star Clusters*.
Talk. "Astronomy on Tap".
Nashville. March 2024.
- ▶ *Binarias de Enanas Blancas*.
Talk. "Colegio Nuestra Señora De La Concepción".
Colombia. March 2024.
- ▶ *Binarias de Enanas Blancas: Sirenas de Ondas Gravitacionales y Laboratorios Estelares*.
Talk. American Association of Variable Star Observers. 2022.
- ▶ *Caminando en la Luna*.
Talk. Planetario de San José UCR. 2021.

Mexico

- ▶ *Observatorios de ondas gravitacionales en la Luna*.
El Centro Astronómico Clavius de la Ibero. 2025.
- ▶ *Carpa Pregúntale a un@ Astrónom@*.
Noche de las Estrellas. 2025.

Texas Tech

- ▶ *A needle in a Haystack: Finding Exotic Binaries in Star Clusters*.
Astronight. YWCA Lubbock. 2022.
<https://slides.com/mmarcano22/ywca2022>

- ▶ *Unveiling Hubble 30th Anniversary Banner.*
Astronight. YWCA Lubbock. 2022.
<https://hubblesite.org/hubble-30th-anniversary/events>
- ▶ *"Hubble's 30th Anniversary/30 aniversario del telescopio Hubble"*
Won a competition to host one of the anniversary banners.
Bilingual talk (Spanish/English) for Astronight event at TTU, Lubbock, TX. 2020
- ▶ *"A tour of the Milky Way/Un Tour por la Vía Lactea"*
Bilingual talk for Astronight event at TTU, Lubbock, TX. 2019
- ▶ *"Dimensionality Reduction applied to Large Spectroscopic Surveys"*
Emmy Noether High School Mathematics Day, Lubbock, TX. 2018

Universidad Autónoma de Santo Domingo (UASD)

- ▶ *Papers Científicos*
Talk. "Como Encontrar Artículos científicos"
Asociación de Estudiantes de Física de la UASD. Abril 2026.
- ▶ *Capacitación de Astronomía para Docentes Universitarios*
Panel. "Avances Recientes en Astronomía"
Universidad Autónoma de Santo Domingo. December 2024.
- ▶ *Capacitación de Astronomía para Docentes Universitarios.*
Talk. "Ciencia Ciudadana: Iniciativas en Astronomía para Involucrar a Estudiantes y Profesores en Investigaciones Astronómicas "
Universidad Autónoma de Santo Domingo. December 2024.
- ▶ *Variabilidad de las binarias de enanas blancas acretantes*
Talk. V Congreso Internacional de la Sociedad Dominicana de Física. Enero 2024.
- ▶ *Astronomía Estelar*
Talk. "Mes de la Astronomía en la República Dominicana".
Universidad Autónoma de Santo Domingo. Summer 2023.

Dominican Society of Amateur Astronomers (Astrodom)

- ▶ *"Observatorio de Ondas Gravitacionales en la Luna"*
Santo Domingo, Dominican Republic. June, 2025
- ▶ *"Una aguja en un pajar: Buscando remanentes estelares en cumulos globulares"*
Santo Domingo, Dominican Republic. 2020
- ▶ *"Galactic Bulge Survey: Fuentes de Rayos X del Bulbo Galactico"*
Santo Domingo, Dominican Republic. 2018
- ▶ *"Estudio de la poblacion de objetos compactos en el cumulo globular NGC 6397 con la unidad de campo integral MUSE"*
Santo Domingo, Dominican Republic. 2016

References

Dr. Liliana Rivera Sandoval

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